

A Theoretical Study of a Ti-Diffused Er:LiNbO₃ Waveguide Laser

Delong Zhang, Caihe Chen, Jiang Li, Guilan Ding, Xiaojun Chen, and Yuming Cui

Abstract— A detailed theoretical procedure of determining the mode sizes of fundamental mode field distribution in a coaxially pumped Ti-diffused Er:LiNbO₃ waveguide laser has been described. The mode sizes, effective refractive indexes, effective pump area, and coupling efficiency as a function of the initial Ti stripe width W were numerically calculated for the waveguide with 95- and 100-nm initial Ti stripe thicknesses, respectively. The results indicated that the threshold pump power is severely affected by the stripe width, while slope efficiency is hardly changed as W ; both show little difference between 95- and 100-nm stripe thickness. In addition, the stimulated emission cross section of Er³⁺ in Er:LiNbO₃ channel waveguide versus the wavelength were calculated directly from its fluorescence spectra using the β - τ method. Subsequently, threshold pump power and slope efficiency were evaluated.

I. INTRODUCTION

RARE-EARTH doping of LiNbO₃ increases significantly the potential of this material, allowing a monolithic integration, and has received much attention in recent years. By combining the excellent laser gain of rare-earth ions with the electrooptic, acoustooptic, and nonlinear optical properties of LiNbO₃, more efficient monolithic laser devices could be realized. Recently, Nd:MgO:LiNbO₃ channel waveguide lasers [1]–[5] and integrated devices including FM mode-locked [6], Q -switched [7], [8], and waveguide lasers have been demonstrated. However, the transition used in Nd³⁺ corresponds to a wavelength of 1.086 μm , which is not well suited for application in optical communication systems. Er³⁺ is the best candidate for providing amplification in the third communication window due to the transition at 1.53 μm . The first CW Ti-diffused Er-diffused LiNbO₃ waveguide laser was reported experimentally in 1991 by Brinkmann *et al.* [9], and in the same year Helmfried *et al.* [10] presented the Er-implanted Ti-diffused LiNbO₃ waveguide amplifier. Then, the experimental results of modelocked Er:Ti:LiNbO₃ waveguide laser were demonstrated by Suche *et al.* [11].

The theoretical study of the amplification or laser oscillation can supply scientific basis for the design and optimization of efficient amplifiers or laser devices. Based on a quasi-two-level model for Er³⁺ ions, an accurate theory has been developed for Er-doped Ti-diffused LiNbO₃ waveguide amplifier pumped by radiation of $\lambda_p = 1.48 \mu\text{m}$ [12]. By varying characteristic

parameters, such as the launched pump power, the polarization and the wavelength of the signal and the pump, the Er concentration profile, and the length of the waveguide, the gain behavior of Er-doped Ti:LiNbO₃ waveguide amplifiers has been studied and discussed in detail. However, the laser oscillation analysis in an Er:Ti:LiNbO₃ waveguide laser has not yet been reported until now. In this paper, we present the theoretical results of Ti-diffused Er:LiNbO₃ waveguide laser. First, by starting with Gaussian/Hermite–Gaussian mode field distributions in the direction of width/depth of the Ti-diffused waveguide and using variational technique, the signal ($\lambda_s = 1.53 \mu\text{m}$), pump ($\lambda_p = 1.477 \mu\text{m}$) mode sizes and effective refractive index versus the stripe width are numerically calculated for the waveguide with 95- and 100-nm initial Ti stripe thickness, respectively. Secondly, the corresponding effective pump area and coupling efficiency between pump beam and laser beam, which are proportional to the threshold pump power and slope efficiency, respectively, are calculated. Thirdly, we calculate the stimulated emission cross section of Er³⁺ in the waveguide directly from its fluorescence spectra prior to the calculation of threshold pump power and slope efficiency. Finally, we summarize the results and finish with some concluding remarks.

II. THEORETICAL DESCRIPTION

A. Variational Analysis

The theoretical analysis of guided mode field distributions for Ti-diffused LiNbO₃ waveguide [13] indicated that the fundamental mode field profile in the direction of the width is well described by symmetric Gaussian function and that the description of the profile in the depth direction can be approximated by a Hermite–Gaussian trial solution. As a result, the normalized mode field can be written as

$$\phi(x, y) = \frac{4y}{\sqrt{\pi W_x W_y^3}} \exp \left[- \left(\frac{x^2}{W_x^2} + \frac{y^2}{W_y^2} \right) \right] \quad (1)$$

where W_x and W_y are the mode sizes. The propagation constant β is given from variational equation

$$\beta^2 = \frac{\iint \phi \nabla^2 \phi \, dx \, dy + k_0^2 \iint n^2 \phi^2 \, dx \, dy}{\iint \phi^2 \, dx \, dy} \quad (2)$$

where ϕ is the guided mode field distribution, k_0 is the wave number in free space. n is the refractive-index profile for a

Manuscript received October 24, 1995; revised June 26, 1996. This work was supported by the Natural Foundation of Tianjin, China.

D. Zhang, C. Chen, G. Ding, and Y. Cui are with the Department of Precision Instrument Engineering, Tianjin University, Tianjin 300072, China.

J. Li and X. Chen are with the Department of Physics, Nankai University, Tianjin 300071, China.

Publisher Item Identifier S 0018-9197(96)07277-6.

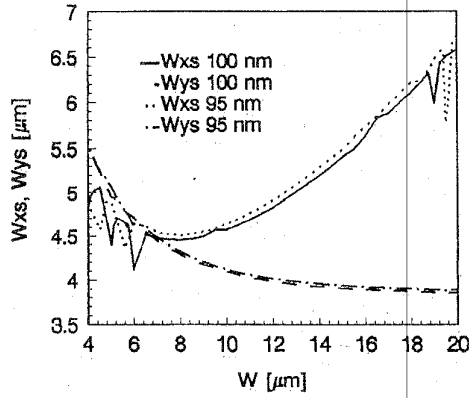


Fig. 1. Laser mode sizes W_{xs} and W_{ys} versus waveguide stripe width for 95- and 100-nm initial Ti stripe thicknesses.

c-cut Ti:LiNbO₃ channel waveguide with the initial width of the Ti stripe W , which can be expressed as

$$n^2(x, y) = \begin{cases} 1 & y < 0 \\ n_b^2 + (n_s^2 - n_b^2)f\left(\frac{y}{d_y}\right)g\left(\frac{2x}{W}\right) & y > 0 \end{cases} \quad (3)$$

in which

$$f\left(\frac{y}{d_y}\right) = \exp\left(-\frac{y^2}{d_y^2}\right) \quad (4)$$

$$g\left(\frac{2x}{W}\right) = \frac{1}{2} \left\{ \operatorname{erf}\left[\frac{W}{2d_x}\left(1 + \frac{2x}{W}\right)\right] + \operatorname{erf}\left[\frac{W}{2d_x}\left(1 - \frac{2x}{W}\right)\right] \right\} \quad (5)$$

where d_x and d_y are diffusion width and depth, respectively, and they can be calculated from the equations

$$d_x = 2\sqrt{D_x t} \quad (6)$$

$$d_y = 2\sqrt{D_y t} \quad (7)$$

where t is the diffusion time, and D_x and D_y are the diffusion coefficients, given by the expressions

$$D_x = D_0^x \exp\left(-\frac{E_0^x}{KT}\right) \quad (8)$$

$$D_y = D_0^y \exp\left(-\frac{E_0^y}{KT}\right) \quad (9)$$

in which D_0^x and D_0^y are the diffusion constants and E_0^x and E_0^y are the activation energies in the x and y direction, respectively, and these data can be obtained from [14]. K is the Boltzmann's constant and T is the diffusion temperature.

In (3), n_b is the refractive index of the substrate, and n_s is the surface index of the waveguide. For c-cut LiNbO₃, there are the relations

$$\begin{aligned} n_{es} &= n_{eb} + \Delta n_e \\ n_{os} &= n_{ob} + \Delta n_o. \end{aligned} \quad (10)$$

Here, the subscripts o and e denote ordinary and extraordinary rays, respectively. According to the dispersion relation of

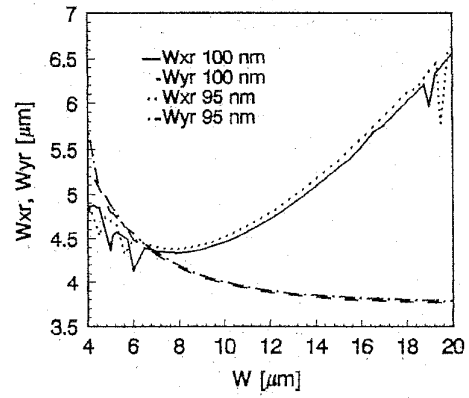


Fig. 2. Pump mode sizes W_{xr} and W_{yr} versus waveguide stripe width for 95- and 100-nm initial Ti stripe thicknesses.

refractive index [15], n_{eb} and n_{ob} can be obtained. The calculation of Δn_e and Δn_o are performed by [14]

$$\Delta n_{e,o} = \left[B_0(\lambda) + B_1(\lambda) \frac{H}{d_{ye,o}} \right] \left(\frac{H}{d_{ye,o}} \right)^{\alpha_{e,o}} \quad (11)$$

for ordinary ray

$$B_0(\lambda) = 0.0653 - 0.0315\lambda + 0.00709\lambda^2$$

$$B_1(\lambda) = 0.478 + 0.464\lambda - 0.348\lambda^2$$

and for extraordinary ray

$$B_0(\lambda) = 0.385 - 0.43\lambda + 0.171\lambda^2$$

$$B_1(\lambda) = 9.13 + 3.85\lambda - 2.49\lambda^2.$$

In (11), H is the initial thickness of the Ti stripe and the fitted constants $\alpha_e = 0.83$, $\alpha_o = 0.53$.

Substituting (1) and (3) into (2), we have

$$\beta^2 = k_0^2 n_b^2 + \frac{4(n_s^2 - n_b^2)k_0^2}{\sqrt{2\pi}W_x} \left(1 + \frac{W_y^2}{2d_y^2} \right)^{-3/2} Q - \frac{1}{W_x^2} - \frac{3}{W_y^2} \quad (12)$$

where

$$Q = \int_0^{+\infty} g\left(\frac{2x}{W}\right) \exp\left(-\frac{2x^2}{W_x^2}\right) dx. \quad (13)$$

Inserting the known values of λ , W , n_b , n_s , d_x , and d_y , the optimum values of W_x and W_y correspond to the large value of β^2 , namely β_m^2 , and then the effective refractive index can be determined from

$$N_{\text{eff}} = \frac{\lambda}{2\pi} \beta_m. \quad (14)$$

B. Laser Oscillation Characteristics

By starting with the rate equation and taking into account the spatial distributions of pump and laser mode, Kubodera and Otsuka [16] have derived the general expressions of threshold pump power and slope efficiency for a coaxially pumped

TABLE I
PARAMETERS USED IN THE CALCULATION OF W_x AND W_y FOR
Er:LiNbO₃ WAVEGUIDE LASER PUMPED BY $\lambda = 1.477 \mu\text{m}$ LIGHT

$\lambda (\mu\text{m})$	1.53		1.477	
H(nm)	95	100	95	100
h(nm/h)	12	12	12	12
T(°C)	1050	1050	1050	1050
n_{e1}	2.1380	2.1380	2.13957	2.13957
$D_0^{s*} (\times 10^{10} \mu\text{m}^2/\text{h})$	0.0135	0.0135	0.0135	0.0135
$D_0^{i*} (\times 10^{10} \mu\text{m}^2/\text{h})$	2.6	2.6	2.6	2.6
$E_0^{s*} (\text{eV})$	2.22	2.22	2.22	2.22
$E_0^{i*} (\text{eV})$	2.77	2.77	2.77	2.77
$d_{xs} (\mu\text{m})$	3.90	4.00	3.90	4.00
$d_{ys} (\mu\text{m})$	4.86	4.98	4.86	4.98
Δn_e	0.0117	0.0122	0.011691	0.01215
n_{e2}	2.1497	2.1502	2.151261	2.15172

waveguide laser. Then, threshold pump power P_{th} and slope efficiency S_e near the threshold can be rewritten as

$$P_{th} = \frac{h\nu_p}{\eta_a \sigma \tau_f} \frac{\delta_1}{2} A_{\text{eff}} \quad (15)$$

$$S_e = \frac{T_2}{\delta_1} \frac{\lambda_p}{\lambda_s} \eta_a \eta_c \quad (16)$$

where δ_1 is the round-trip cavity loss, h is the Planck's constant, η_a is the waveguide absorption efficiency for pump light, σ is the stimulated emission cross section of Er³⁺, and τ_f is the fluorescence lifetime. A_{eff} is the effective pump area and is given by

$$A_{\text{eff}} = \left[l \int_{\text{cavity}} s_{01}(x, y, z) r_0(x, y, z) dx dy dz \right]^{-1} \quad (17)$$

Here, l is the cavity length and $s_{01}(x, y, z)$ is the normalized signal mode intensity distribution. For the c-cut Ti:LiNbO₃ waveguide, it can be given by (1) as

$$s_{01}(x, y, z) = \frac{16y^2}{\pi W_{xs} W_{ys}^3 l} \exp \left[-2 \left(\frac{x^2}{W_{xs}^2} + \frac{y^2}{W_{ys}^2} \right) \right] \quad (18)$$

$r_0(x, y, z)$ is the normalized distribution function of the pump rate and can be written as

$$r_0(x, y, z) = \frac{16y^2}{\pi W_{xr} W_{yr}^3 l} \exp \left[-2 \left(\frac{x^2}{W_{xr}^2} + \frac{y^2}{W_{yr}^2} \right) \right] \quad (19)$$

In the above two expressions, W_{is} and W_{ir} ($i = x, y$) represent the signal and pump mode sizes, respectively. The z -dependent variations were reasonably neglected. The waveguide laser was assumed to be pumped coaxially. Both the incident pump beam and signal beam are supposed to be waveguided completely in a fundamental mode.

In (16), T_2 is the transmittance of the output mirror at laser wavelength, λ_p/λ_s is the ratio of pump-to-signal wavelength. η_c is the coupling efficiency between pump mode and signal mode and it can be expressed as

$$\eta_c = \frac{I_1^2}{I_2} \quad (20)$$

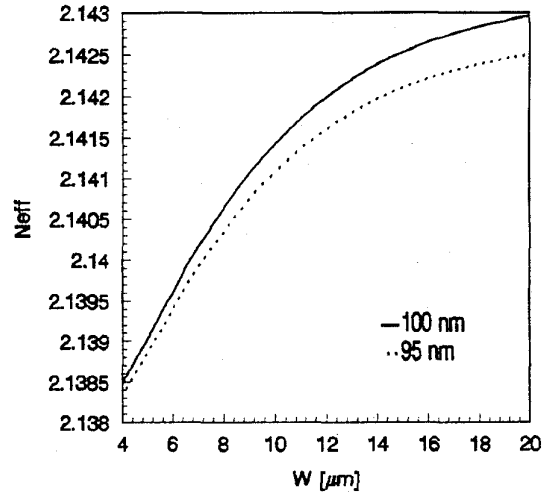


Fig. 3. Variation of effective refractive index with waveguide stripe width for 95- and 100-nm initial Ti stripe thickness.

where

$$I_1 = \int_{\text{cavity}} s_{01}(x, y, z) r_0(x, y, z) dx dy dz \quad (21)$$

$$I_2 = \int_{\text{cavity}} s_{01}^2(x, y, z) r_0(x, y, z) dx dy dz \quad (22)$$

Substituting (18) and (19) into (17) and (20)–(22) numerically, then one finds

$$A_{\text{eff}} = \frac{\pi}{12} \frac{\sqrt{(W_{xs}^2 + W_{xr}^2)(W_{ys}^2 + W_{yr}^2)^5}}{W_{ys}^2 W_{yr}^2} \quad (23)$$

$$\eta_c = \frac{3}{5} \frac{W_{xs} W_{ys}^3 \sqrt{(W_{xs}^2 + 2W_{xr}^2)(W_{ys}^2 + 2W_{yr}^2)^7}}{(W_{xs}^2 + W_{xr}^2)(W_{ys}^2 + W_{yr}^2)^5} \quad (24)$$

Inserting the previous calculated values of W_{is} and W_{ir} ($i = x, y$), one can obtain the corresponding values of A_{eff} and η_c , which are proportional respectively to P_{th} and S_e . Thus, it is important to obtain low values of A_{eff} and high values of η_c in order to minimize the threshold for pumping and to increase the slope efficiency.

III. RESULTS AND DISCUSSIONS

A. Mode Sizes and Effective Refractive Index

According to (12), C program was employed to calculate the mode sizes W_x and W_y . The related parameters used in this calculation are summarized in Table I. In Table I, h is the diffusion rate and the subscript e represents the extraordinary ray. The calculations are performed for the initial Ti stripe thicknesses of 95 and 100 nm in the wavelengths of 1.53 and 1.477 μm . The variations of W_x and W_y with the waveguide stripe width W in the range of 4–20 μm are plotted in Figs. 1 and 2. The effective refractive index as a function of waveguide width for 95- and 100-nm Ti stripe thicknesses are shown in Fig. 3.

One can see that Fig. 1 has the similar characteristic as Fig. 2. W_y decreases as the stripe width is increased and then

TABLE II
PARAMETERS TAKEN FROM [9]: CALCULATED THRESHOLD PUMP POWER AND SLOPE EFFICIENCY FOR C-CUT Er:LiNbO₃ WAVEGUIDE LASER

$\lambda_p(\mu m)$	1.477	$d_{ys}(\mu m)$	7.077	$W_{yr}(\mu m)$	5.0815
$\lambda_s(\mu m)$	1.532	$d_{yp}(\mu m)$	5.6743	$A_{eff}(\mu m^2)$	49.566
$\nu_p(\times 10^{14} Hz)$	2.031	n_{ebi}	2.1380	η_c	77.28 %
$T(^{\circ}C)$	1060	n_{esi}	2.147439	δ_1	0.68
$t(h)$	9	n_{ebp}	2.13957	η_a	14.57 %
$H(nm)$	95	n_{esp}	2.148967	$\tau_f(ms)$	2.9
$W(\mu m)$	7	$W_{xs}(\mu m)$	5.1184	$\sigma(\times 10^{-36} m^2)$	6.8
$d_{xs}(\mu m)$	4.4708	$W_{ys}(\mu m)$	5.2332	T_1	8 %
$d_{xp}(\mu m)$	4.4708	$W_{yp}(\mu m)$	4.986	T_2	0.5 %
$l(mm)$	10.5	$P_{th}(mW)$	7.90(8.0)	S_c	0.08 %(0.4 %)

Values in the parentheses are experimental results presented in [9].

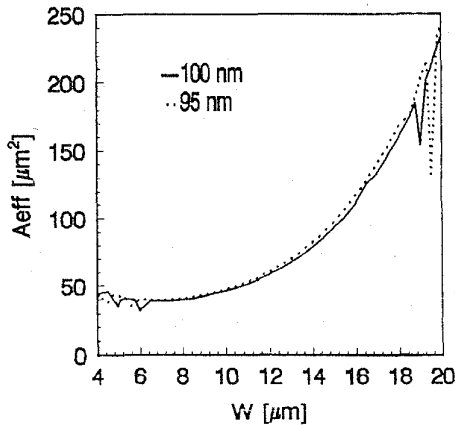


Fig. 4. Effective pump area versus waveguide stripe width for 95- and 100-nm initial Ti stripe thickness.

tends to be constant above 17 μm . But W_{xs} , for the stripe width above 8 μm , increases gradually and then rapidly. Moreover, the similar feature is also observed for the waveguide with 95- and 100-nm stripe thicknesses, respectively.

The effective refractive indexes N_{eff} increase steadily with a rise in stripe width for the waveguide with both 95- and 100-nm initial thickness, as shown in Fig. 3, and the latter increases more rapidly than the former. The values of N_{eff} for the two thicknesses are larger than that of n_b (2.1380) and smaller than that of n_s (2.1502 for 100 nm and 2.1497 for 95 nm). This is consistent with the waveguide theory.

B. Effective Pump Area and Coupling Efficiency

From Figs. 1 and 2, effective pump area A_{eff} and coupling efficiency η_c can be readily calculated. The results are shown in Figs. 4 and 5, respectively. We can see from Fig. 4 that A_{eff} doesn't change essentially for W below 9 μm and then increases strongly with increasing W . In Fig. 5, η_c decreases slightly as W is increased from 6 μm , indicating that η_c can not influence significantly slope efficiency S_c . Furthermore, no obvious difference is observed from both A_{eff} and η_c for the waveguide with 95- and 100-nm stripe thickness, respectively, showing that the initial Ti stripe thickness may give a little influence on the threshold pump power and slope efficiency.

Thus, we may conclude that the threshold pump power is severely affected by the stripe width through effective pump

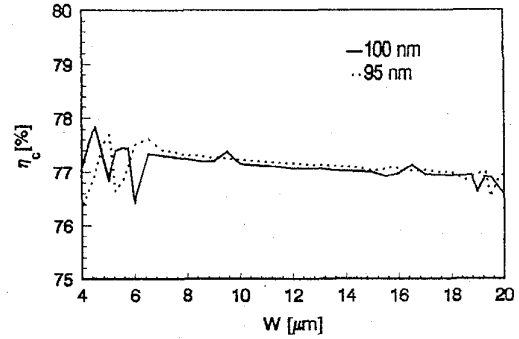


Fig. 5. Coupling efficiency versus waveguide stripe width for 95- and 100-nm initial Ti stripe thickness.

area and hardly by the initial stripe thickness. The appropriate stripe width for the more efficient laser oscillation is about 4–9 μm , at which the waveguide not only shows low loss [17], but also supports merely fundamental mode [18]. On the contrary, the slope efficiency shows little dependence on either stripe width or initial stripe thickness through the coupling efficiency.

C. Threshold Pump Power and Slope Efficiency

This section reports the first calculated results of stimulated emission cross section of Er^{3+} in waveguide with 6 μm stripe width and then gives theoretical values of threshold pump power and slope efficiency on the basis of [9]. This result is also compared with the experimental values. The parameters presented by [9] are collected in Table II.

According to [9], the waveguide was fabricated by a standard Ti indiffusion (1060 $^{\circ}C$, 9 hr) of a 95-nm-thick, 7- μm -wide Ti stripe. At this diffusion condition, diffusion width and depth, refractive-index change, and waveguide-surface refractive index can be calculated in order and shown in Table II, where n_{ebi} and n_{esi} ($i = s, p$) denote substrate and waveguide-surface refractive index, respectively for laser or pump mode for extraordinary ray. As result of $W_{xs} = 5.1184 \mu m$, $W_{ys} = 5.2332 \mu m$, $W_{xp} = 4.986 \mu m$, and $W_{yp} = 5.0815 \mu m$, which give $A_{eff} = 49.566 \mu m^2$ and $\eta_c = 77.28\%$. At $\lambda = 1.477 \mu m$, the absorption coefficient of the waveguide was found to be $\sim 0.15 \text{ cm}^{-1}$ [9], then one can find that the absorption efficiency of the waveguide for the pump light is about 14.57%. The round-trip cavity loss δ_1 is calculated on the basis of the absorption coefficient ($\sim 0.5 \text{ cm}^{-1}$) of

the waveguide and the transmittance of the input and output mirrors for the laser.

The stimulated emission cross section σ is one of the key parameters in determining the threshold pump power as shown in (15). The preliminary results have been reported by Dinand and Sohler [12] using the relationship between absorption and emission cross section. Nevertheless, the calculation was based on the assumption of homogeneous Er-doped bulk LiNbO₃ crystal and the Er concentration was assumed as the surface concentration. Moreover, only σ -polarization cross section as a function of wavelength was issued. Here, we report the completed results of stimulated emission cross section starting directly with the fluorescence spectra using another method, which is referred to as " β - τ " method [19], [20]. According to Krupke [20], the cross section for fluorescence emission at wavelength λ can be expressed as

$$\sigma_{ij}(\lambda) = \frac{\lambda^5}{8\pi c n^2} \frac{\beta_{ij}}{\tau_i} \frac{I(\lambda)}{\int \lambda I(\lambda) d\lambda} \quad (25)$$

where $I(\lambda)$ is the fluorescence intensity at wavelength λ , β_{ij} is the fluorescent branching ratio of the $i \rightarrow j$ transition, n is the refractive index of the host, τ_i is the fluorescence lifetime of the upper state i , and c is the velocity of light in free space. The fluorescence spectra for c-cut Er-diffused LiNbO₃ waveguide with stripe width of 6 μm have been measured by Brinkmann *et al.* [21] for π and σ polarization, which correspond to polarization parallel and perpendicular to the optical axis (i.e., c-axis), respectively. For Er³⁺ $^4I_{13/2} \rightarrow ^4I_{15/2}$ transition, $\beta_{ij} = 1$ and $\tau_i = \tau_f = 2.9$ ms. Fig. 6 presents the calculated results for π and σ polarizations. In comparison with the result of Dinand and Sohler [12], our σ -polarization spectrum has a similar feature as that of Dinand and Sohler: the maximum emission cross section at 1.532 μm in this spectrum is $1.82 \times 10^{-24} \text{ m}^2$, and figures about $2.44 \times 10^{-24} \text{ m}^2$ have been presented by Dinand and Sohler. Reference [9] revealed that laser oscillation operates at $\lambda = 1.532 \mu\text{m}$ and its polarization parallels to the optical axis. One can obtain the cross section of $6.8 \times 10^{-25} \text{ m}^2$ from the π -polarization spectrum as shown in Fig. 6 (the maximum cross section $1.1 \times 10^{-24} \text{ m}^2$ corresponds to $\lambda = 1.535 \mu\text{m}$).

If we now use (15) to calculate P_{th} with the known parameters, we find $P_{th} = 7.9$ mW, in every agreement with the measured value of 8 mW. From (16), we find that S_e should be 0.08%, which is five times lower than the experimental value of 0.4%. This disagreement is not yet fully understood.

IV. CONCLUSION

In this paper, we have presented a detailed procedure of determining the mode sizes of the fundamental guided mode in a coaxially pumped Ti-diffused Er:LiNbO₃ waveguide laser using a variational method. The effective pump area and coupling efficiency, which are proportional to threshold pump power and slope efficiency, respectively, are calculated as a function of Ti stripe width for 95- and 100-nm initial Ti stripe thickness. The calculated results indicated that the stripe

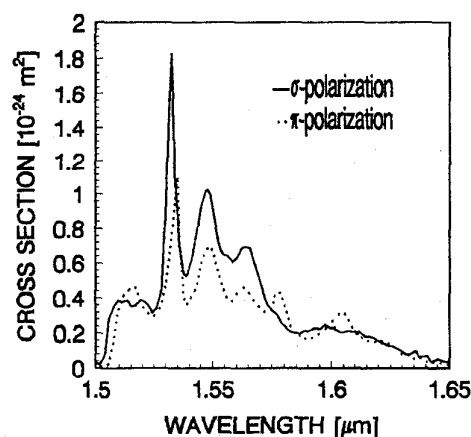


Fig. 6. Wavelength-dependent stimulated emission cross sections of Er:LiNbO₃ waveguide for π and σ polarizations.

width can give significant influence on the threshold pump power for the width above 9 μm , while it barely alters the slope efficiency. Moreover, both threshold pump power and slope efficiency are essentially unchanged between 95- and 100-nm initial stripe thicknesses. The appropriate stripe width for more efficient laser oscillation is 4–9 μm , at which the waveguide laser not only shows low loss and threshold pump power through effective pump area, but also supports only the fundamental mode. In addition, the stimulated emission cross section of Er³⁺ in Er:LiNbO₃ waveguide has been directly calculated from its fluorescence spectra using the β - τ method. By applying this result to the calculation of the threshold pump power, an excellent agreement between theory and experiment is achieved.

REFERENCES

- [1] E. Lallier, J. P. Pocholle, M. Papuchon, C. Grezes-Besset, E. Pelletier, M. De Micheli, M. J. Li, Q. He, and D. B. Ostrowsky, "Laser oscillation of single-mode channel waveguide in Nd:MgO:LiNbO₃," *Electron. Lett.*, vol. 25, p. 1491, 1989.
- [2] E. Lallier, J. P. Pocholle, M. Papuchon, M. De Micheli, M. J. Li, Q. He, D. B. Ostrowsky, C. Grezes-Besset, and E. Pelletier, "Efficient Nd:MgO:LiNbO₃ waveguide laser," *Electron. Lett.*, vol. 26, p. 927, 1990.
- [3] —, "Nd:MgO:LiNbO₃ waveguide laser and amplifier," *Opt. Lett.*, vol. 15, p. 682, 1990.
- [4] —, "Nd:MgO:LiNbO₃ channel waveguide laser devices," *IEEE J. Quantum Electron.*, vol. 27, p. 618, 1991.
- [5] R. Brinkmann, W. Sohler, H. Suche, and C. Wersig, "Fluorescence and laser operation in single mode Ti-diffused Nd:MgO:LiNbO₃ waveguide structures," *IEEE J. Quantum Electron.*, vol. 28, p. 466, 1992.
- [6] E. Lallier, J. P. Pocholle, M. Papuchon, Q. He, M. De Micheli, D. B. Ostrowsky, C. Grezes-Besset, and E. Pelletier, "Integrated Nd:MgO:LiNbO₃ FM mode-locked waveguide laser," *Electron. Lett.*, vol. 27, p. 936, 1991.
- [7] E. Lallier, J. P. Pocholle, M. Papuchon, Q. He, M. De Micheli, and D. B. Ostrowsky, "Integrated Q-switched Nd:MgO:LiNbO₃ waveguide laser," *Electron. Lett.*, vol. 28, p. 1428, 1992.
- [8] E. Lallier, D. Papillon, J. P. Pocholle, M. Papuchon, M. De Micheli, and D. B. Ostrowsky, "Short pulse, high power Q-switched Nd:MgO:LiNbO₃ waveguide laser," *Electron. Lett.*, vol. 29, p. 175, 1993.
- [9] R. Brinkmann, W. Sohler, and H. Suche, "Continuous-wave Erbium-diffused LiNbO₃ waveguide laser," *Electron. Lett.*, vol. 27, p. 415, 1991.
- [10] S. Helmfrid, G. Arvidsson, J. Webjörn, M. Linnarsson, and T. Pihl, "Stimulated emission in Er:Ti:LiNbO₃ waveguides close to 1.53 μm transition," *Electron. Lett.*, vol. 27, p. 913, 1991.

- [11] H. Suche, L. Baumann, D. Hiller, and W. Sohler, "Modelocked Er:Ti:LiNbO₃ waveguide laser," *Electron. Lett.*, vol. 29, p. 1111, 1993.
- [12] M. Dinand and W. Sohler, "Theoretical modeling of optical amplification in Er-doped Ti:LiNbO₃ waveguides," *IEEE J. Quantum Electron.*, vol. 30, p. 1267, 1994.
- [13] N. H. Zhu, Y. K. Lin, Z. D. Wu, and W. G. Lin, "Analytical expressions of the guided mode field distribution: Analysis and comparison," *IEEE J. Quantum Electron.*, vol. 28, p. 1424, 1992.
- [14] S. Fouchet, A. Carencio, C. Daguet, R. Guglielmi, and L. Riviere, "Wave-length dispersion of Ti induced refractive index change in LiNbO₃ as a function of diffusion parameters," *J. Lightwave Technol.*, vol. 5, p. 700, 1987.
- [15] M. V. Hobden and J. Warner, "The temperature dependence of the refractive indices of pure Lithium Niobate," *Phys. Lett.*, vol. 22, p. 243, 1966.
- [16] K. Kubodera and K. Otsuka, "Single-transverse-mode LiNdP₄O₁₂ slab waveguide laser," *J. Appl. Phys.*, vol. 50, p. 653, 1979.
- [17] M. Fukuma and J. Noda, "Optical properties of titanium-diffused LiNbO₃ strip waveguides and their coupling-to-a-fiber characteristics," *Appl. Opt.*, vol. 19, p. 591, 1980.
- [18] N. H. Zhu, X. S. Zheng, Y. K. Lin, and X. Yu, "Calculation of the field distribution of a Ti:LiNbO₃ optical waveguide and its applications," *Opt. Quantum Electron.*, vol. 24, p. 737, 1992.
- [19] B. F. Aull and H. P. Jenssen, "Vibronic interactions in Nd:YAG resulting in nonreciprocity of absorption and stimulated emission cross sections," *IEEE J. Quantum Electron.*, vol. QE-18, p. 925, 1982.
- [20] W. F. Krupke, M. D. Shinn, J. E. Marion, J. A. Caird, and S. E. Stokowski, "Spectroscopic, optical, and thermomechanical properties of neodymium- and chromium-doped gadolinium scandium gallium garnet," *J. Opt. Soc. Amer. B*, vol. 3, p. 102, 1986.
- [21] R. Brinkmann, W. Sohler, and H. Suche, "Absorption, fluorescence and stimulated emission in Ti-diffused Er:LiNbO₃ waveguides," *SPIE Physical Concepts of Materials for Novel Optoelectronic Device Applications II: Device Physics and Applications*, vol. 1362, p. 377, 1990.

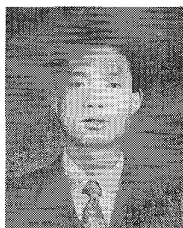
Caihe Chen, photograph and biography not available at the time of publication.

Jiang Li, photograph and biography not available at the time of publication.

Guilan Ding, photograph and biography not available at the time of publication.

Xiaojun Chen, photograph and biography not available at the time of publication.

Yuming Cui, photograph and biography not available at the time of publication.



Delong Zhang was born in Tianjin, China, in 1965. He received the B.S. and M.S. degrees from the Normal University of Tianjin, China, in 1986 and 1989, respectively, and the Ph.D. degree from Nankai University, Tianjin, China, in 1994.

While at Normal University, his research work was in the field of interactions between ion beams and solid surfaces. While at Nankai University, his research concerned lattice vibrational spectra and high-pressure Raman spectra. After receiving the Ph.D. degree in 1994, he joined the Department of

Precision Instrument Engineering, Tianjin University, Tianjin, China, where his research interests are in the field of integrated optics and devices.